# Unit 2: Market Equilibrium and Elasticity of Demand

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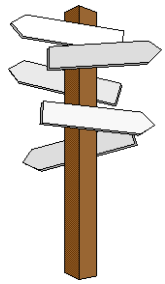
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# Must Read Before You Begin



In the last unit we introduced demand and supply, whose interaction determines the price in a market. In this unit we will give a complete picture of the price mechanism. We will introduce the market equilibrium and also look at some cases in which a market is prevented from reaching its equilibrium. Then we will focus on the demand side by studying the demand elasticity and the consumer surplus. Detailed analysis of the supply side will be done in the next unit.

## Topics

There are four main topics in Unit 2.

### Topic 2.1: Price Mechanism – Demand, Supply and Market (Part II)

This section is a continuation of Topic 1.3 of Unit 1. We will put market demand and market supply together and see how market equilibrium will be reached. We will also study the changes in market equilibrium brought about by changes in the determinants of demand and supply.

### Topic 2.2: Consumer Surplus and producer Surplus

In this section we introduce the concepts of consumer surplus, which is the gain consumers derive from their purchase of a good or service, and producer surplus, which is the gain producers derive from their production and sale of a good or service.

### Topic 2.3: Price Ceiling and Price Floor

In this section we study the two cases in which the market is prevented from reaching its equilibrium. This occurs when the government imposes either a price ceiling or a price floor on a good.

### Topic 2.4: Elasticity of Demand

In this topic we introduce the following elasticities of demand:

- price elasticity;
- income elasticity; and
- cross elasticity.

We will also analyse the implications of price elasticity of demand on the total revenue earned by the supplier of a good and the tax revenue collected by the government.

## Your Objectives

When you have worked through this unit you should be able to complete the following objectives.

1. Define market equilibrium and illustrate it in a graph.
2. Analyse and illustrate in a diagram how the market equilibrium will be affected by changes in any of the determinants of demand or supply.
3. Define consumer surplus and producer surplus and illustrate them in a diagram.
4. Analyse the effects of a price ceiling and a price floor on a market.
5. Define price elasticity of demand and identify its four determinants.
6. Explain the relationship between price elasticity and total revenue and apply it to the analysis of real life problems.
7. Define income elasticity of demand and use it to differentiate between normal and inferior goods.
8. Define cross elasticity of demand and use it to differentiate between substitutes and complements.

## Self Exercises

There are seven (7) self exercises in this unit.

Let us continue with our discussion of price mechanism from the last unit.

# 2.1: Price Mechanism – Demand, Supply and Market (Part II)

This section is a continuation of Topic 1.3 of Unit 1. We will put market demand and market supply together and see how market equilibrium will be reached. We will also study the changes in market equilibrium brought about by changes in the determinants of demand and supply.

## 2.1.1: Market Equilibrium

An equilibrium is a situation in which there is no inherent tendency to change. In a market, equilibrium is reached when the quantity demanded and the quantity supplied of a good are equal at the given price. In a demand-supply diagram as illustrated in Figure 2.1, a market equilibrium is the intersection of the demand curve and the supply curve. At this intersection (point **e**), the price is the **equilibrium price** whereas the quantity bought and sold (or transacted) is the **equilibrium quantity**.

## Adjustment to Equilibrium

In a market economy where the price is allowed to adjust freely and the economic agents including consumers and producers are free to determine their buying and selling plans, a market will move to its equilibrium. Price plays a crucial role in the following two ways.

1. Consumers adjust their buying plans (quantity demanded) and producers adjust their selling plans (quantity supplied) according to the price of the good concerned.

2. When the quantity demanded and the quantity supplied are not the same, price adjusts to balance them.

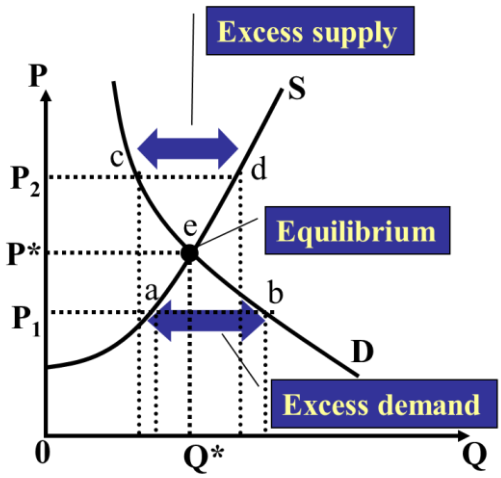



**Figure 2.1**
Market Equilibrium

In Figure 2.1, **P*** and **Q*** represent the equilibrium price and equilibrium quantity respectively. What will happen if the market is away from the equilibrium (i.e. when quantity demanded does not equal quantity supplied)? As has been mentioned already, the price will adjust.

There are two directions in which the price can adjust.

1. Price adjusts upwards when there is an excess demand.

   - When the price is below the equilibrium level, say, at $\mathbf{P_1}$ in Figure 2.1, quantity demanded exceeds quantity supplied. The difference represented by **ab** is known as excess demand or shortage.

   - In response to the shortage, the market price rises inducing producers to increase their output (represented by a movement along the market supply curve upward from **a** towards **e**). At the same time, the consumers reduce their purchase in response to the higher price (moving along the market demand curve upward from **b** towards **e**). The price will continue to rise until the buying plan and selling plan of these economic agents balance with each other.

- Price will settle at the equilibrium level **P*** at which quantity demanded equals quantity supplied.

2. Price adjusts downwards when there is an excess supply.

   - When the price is above the equilibrium level, say, at $\mathbf{P_2}$ in Figure 2.1, quantity supplied exceeds quantity demanded. The difference represented by **cd** is known as **excess supply** or **surplus**.

   - In response to the surplus, the market price drops making producers reduce their output (represented by a movement along the market supply curve downward from **d** towards **e**). At the same time, the consumers increase their purchase in response to the lower price (moving along the market demand curve downward from **c** towards **e**). The price will continue to drop until the buying plan and selling plan of these economic agents balance with each other.

   - Price will settle at the equilibrium level **P*** at which quantity demanded equals quantity supplied.

## 2.1.2: Changes in Equilibrium

When the demand and supply of a good remain unchanged, there is no inherent tendency for the market equilibrium to change. However, it is unlikely for these two forces to remain the same for long. Any changes in their determinants introduced in Topic 1.3 of Unit 1 will disturb the original equilibrium and lead to a series of adjustments in both the price and quantity of the good concerned.

### A Change in Demand

Here we assume that supply remains unchanged. Depending on the direction of the change in demand, there are two possible outcomes, both illustrated in Figure 2.2.

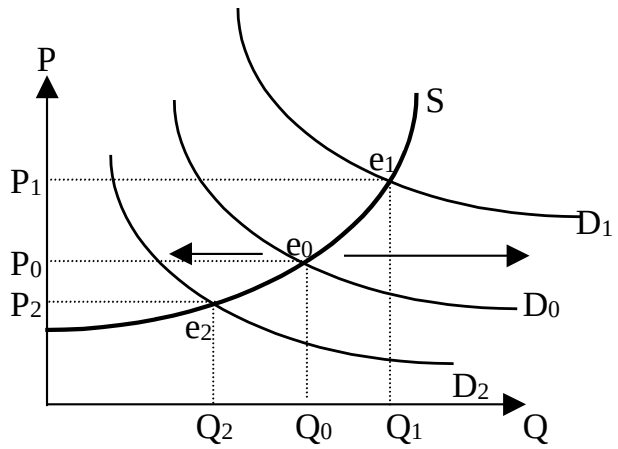



**Figure 2.2**
Adjustment to Changes in Demand.

1. An increase in demand

   In Figure 2.2, as the demand increases from $\mathbf{D_0}$ to $\mathbf{D_1}$, there will be an excess demand at the original equilibrium price $\mathbf{P_0}$. Price will be pushed up until quantity demanded and quantity supplied are equal at a higher price level $\mathbf{P_1}$. This higher price induces the producers to produce and sell more of the good and therefore the quantity transacted increases to $\mathbf{Q_1}$. Equilibrium changes from $\mathbf{e_0}$ to $\mathbf{e_1}$ in the diagram.

2. A decrease in demand

   When the demand decreases from $\mathbf{D_0}$ to $\mathbf{D_2}$, there will be an excess supply at the original equilibrium price $\mathbf{P_0}$. Price will be forced down until quantity demanded and quantity supplied are equal at a lower price level $\mathbf{P_2}$. This lower price reduces the production of the good and therefore the quantity transacted drops to $\mathbf{Q_2}$. Equilibrium changes from $\mathbf{e_0}$ to $\mathbf{e_2}$ in the diagram.

**When there is a change in demand while supply remains the same, equilibrium price and equilibrium quantity change in the same direction.**

## A Change in Supply

Here we assume that demand remains unchanged. Depending on the direction of the change in supply, there are two possible outcomes, both illustrated in Figure 2.3.

1. An increase in supply

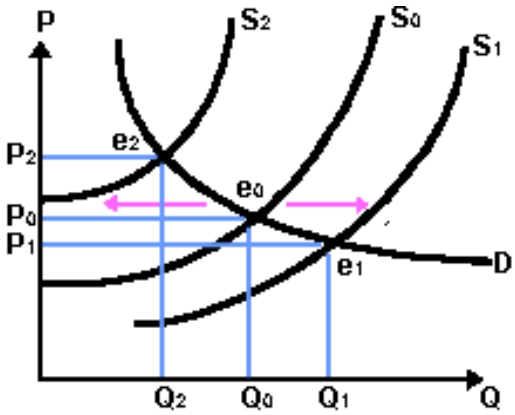



**Figure 2.3**
Adjustment to Changes in Supply.

   When the supply of a good increases from $\mathbf{S_0}$ to $\mathbf{S_1}$, say, due to technological progress, the equilibrium price of the good falls from $\mathbf{P_0}$ to $\mathbf{P_1}$, while the equilibrium quantity increases from $\mathbf{Q_0}$ to $\mathbf{Q_1}$.

2. A decrease in supply

   When the supply of a good decreases from $\mathbf{S_0}$ to $\mathbf{S_2}$, say, due to an increase in factor prices, the equilibrium price of the good rises from $\mathbf{P_0}$ to $\mathbf{P_2}$, while the equilibrium quantity drops from $\mathbf{Q_0}$ to $\mathbf{Q_2}$.

**When there is a change in supply while demand remains the same, equilibrium price and equilibrium quantity change in opposite directions.**

## A Change in Both Demand and Supply

As a detailed explanation is available in the textbook reading, we will only provide a summary of the possible outcomes in the following table.

| Changes in demand and supply | Supply increases | Supply decreases |
|---|---|---|
| **Demand increases** | Price uncertain<br>Quantity increases | Price increases<br>Quantity uncertain |
| **Demand decreases** | Price decreases<br>Quantity uncertain | Price uncertain<br>Quantity decreases |

**Table 2.1**
Adjustment in Equilibrium to Changes in Both Demand and Supply

In Table 2.1, depending on the combination of the changes in demand and supply, sometimes we cannot decide readily how the equilibrium price or quantity will adjust. Can you tell what additional information is needed in order to solve the problem for each of the four uncertain outcomes?

In the following self exercise, you can test your understanding of the price mechanism in various cases, most of which are real life examples.

**Self Exercise 1**

Adjustment of Market Equilibrium



In this exercise we assume a downward sloping demand curve and an upward sloping supply curve and that other things remain unchanged apart from the specific events. For each of the events, indicate whether the equilibrium price and quantity of the good (printed in **bold**) increases or decreases by putting a '✓' in the appropriate box under price **and** quantity. As shown in Table 2.1, when both demand and supply of the good change at the same time, we may not be able to decide readily how the equilibrium price or quantity will adjust. If this is the case, put a '✓' in the box under "Uncertain."

| Event | Price | | | Quantity | | |
|---|---|---|---|---|---|---|
| | **Increase** | **Decrease** | **Uncertain** | **Increase** | **Decrease** | **Uncertain** |
| 1. The increase in tariff (an import tax) rate on the imports of **wine**. | | | | | | |
| 2. The previously illegal sales of a **drug** are now legalised. | | | | | | |
| 3. Medical research has found that **kiwifruit** is a cure of cancer. | | | | | | |
| 4. How would a good harvest of potatoes affect the market of **French fries**? | | | | | | |
| 5. Over time the technology of computer production has improved substantially while the use of **computers** has become more common. | | | | | | |

A response to this self exercise is on page 31.

# 2.2: Consumer Surplus and Producer Surplus

In this section we introduce the concepts of consumer surplus, which is the gain consumers derive from their purchase of a good or service, and producer surplus, which is the gain producers derive from their production and sale of a good or service. Both of them are examples of economic surplus that was introduced in Unit 1, which is the excess of benefit over cost to those economic agents who take part in market transactions.

## 2.2.1: Consumer Surplus

To understand consumer surplus, it is best to start with your own experience. Have you ever paid more than what you are willing to pay for a certain good or service? If you are a rational consumer, the answer should be 'no'. On the other hand, quite often you pay less than the amount that you are willing to pay. That happens when you have a bargain.

**Consumer surplus is equal to the benefit a consumer can derive from the good minus the cost he or she incurs. In the case of a good purchased on the market, consumer surplus is the extra amount that a consumer is willing to pay over and above what he or she actually pays.**

Let us revisit the concept of marginal benefit introduced in Unit 1. For each additional unit of a good you consider buying, the marginal benefit you can derive from the consumption of this extra unit determines the maximum amount you are willing to pay for it. This maximum amount is called the ***reservation price***. You will buy the good only if the market price is less than or equal to this reservation price. For example, if the price of a cup of coffee is $10. If the marginal benefit you can get from this cup of coffee is $8, your reservation price will be $8 and you will not pay $10 for it. If your reservation price is $10, you may or may not buy it in reality. But in economic analysis, we assume that you will buy it although you will not get any consumer surplus in this case. If your reservation price is $12, you will be happy to pay $10 to get that cup of coffee and gain $2 of consumer surplus.

If you buy more than one unit of a good, your total consumer surplus is the summation of the consumer surplus you gain from each of the units you buy at the market price. Refer to Example 1 under the section Something to Think About on p.10 of Unit 1. If the columns of total benefit and marginal benefit describe your preference, how many units of the good will you purchase and how much will be your consumer surplus? I have included a table on the next page that shows the original columns and three more to illustrate how the consumer surplus (CS) can be derived and the resulting rational decision of yours.

| Q | Marginal cost ($) | Total benefit ($) | Marginal benefit ($) | Increase in CS ($) | Total CS ($) | Buy or not? |
|---|---|---|---|---|---|---|
| 1 | 10 | 40 | 40 | 30 | 30 | √ |
| 2 | 10 | 60 | 20 | 10 | 40 | √ |
| 3 | 10 | 75 | 15 | 5 | 45 | √ |
| 4 | 10 | 85 | 10 | 0 | 45 | √ |
| 5 | 10 | 90 | 5 | -5 | 40 | × |

This example is a *discrete* case where the quantity of a good is not divisible, such as the purchase of coffee per cup. For this type of questions, doing the analysis in a table as shown above will be better than using a diagram. The third last column in the table shows how much economic surplus each additional unit of the good will bring. The second last column shows the total consumer surplus a certain quantity of the good will bring to the consumer. At the price of $10, the consumer will only buy 4 units to maximize the consumer surplus, which is equal to $45 in this case.

Buying more than that will make the consumer worse off, as shown by the decrease in consumer surplus.

The discrete case is most relevant to the analysis of individual consumers. In theoretical analysis, however, we may use the *continuous* case in which the quantity of a good is assumed to be perfectly divisible, i.e. you can buy any fraction of a unit of the good by paying the corresponding amount. In such a case, a diagram such as Figure 2.4 will be useful to illustrate consumer surplus.

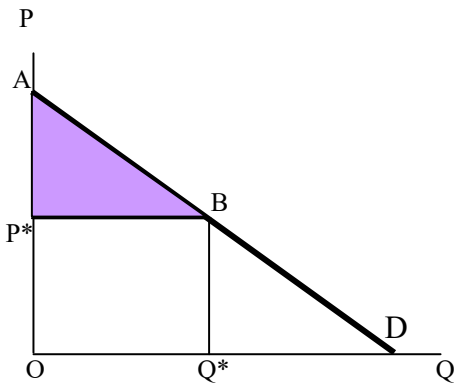



**Figure 2.4**
Consumer Surplus

In Topic 1.3.1, we mentioned two ways of interpreting the demand curve. The second interpretation is that the demand curve gives us the maximum price that a consumer is willing to pay for a given quantity of the good, i.e. the reservation price. Therefore, the area under the demand curve, given by the area **OABQ***, represents the maximum amount that the consumer is willing to pay for **Q*** of the good. It also reflects the total benefit he/she can get out of that quantity. The actual amount that the consumer has to pay is given by the area **OP*BQ***. The difference between these two areas is the consumer surplus, represented by the shaded area **P*AB**. It is the benefit that a consumer can derive from the good but does not have to pay for.

In the case of a market where there are a large number of consumers, the *continuous* case is more applicable as the quantity is usually expressed in thousands or millions of units and therefore it is obviously divisible, i.e. you can buy any fraction of a unit (in thousands or millions) of the good by paying the corresponding amount. If the demand curve in Figure 2.4 denotes the market demand, then the consumer surplus gained by all the consumers who buy the good at the market price P* can be illustrated in the same way as explained above.

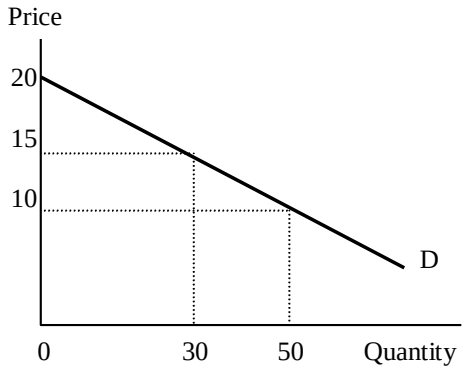
In the following self exercise, try to work out the numerical value of the consumer surplus.



**Self Exercise 2**

Consumer Surplus

Answer the following four questions with reference to the diagram below. D is the demand curve that shows David's demand for good X.




1. When the price equals 15, what is the maximum amount that David is willing to pay for the 30 units of X?

   _____

2. What is David's consumer surplus when the price equals 15?

   _____

3. What is David's consumer surplus when the price is reduced to 10?

   _____

4. What can you conclude from the above analysis?

5. The lower the price of a good, the _____ (larger/smaller) the consumer surplus.

The response to this self exercise is on page 32.

## 2.2.2: Producer Surplus

If you have the experience of producing and selling something at a price, you will probably remember that you usually sell it at a price above the minimum level acceptable to you. Here the benefit from the transaction is the price you charge and the cost is the minimum price acceptable to you.

**Producer surplus is equal to the benefit a producer can derive from the production and sale of a good minus the cost he or she incurs. In other words, it is the extra amount that a producer receives over and above the minimum**

**amount that is needed to induce him/her to produce and supply a given quantity of the product.**

Let us revisit the concept of marginal cost introduced in Unit 1. The minimum amount that is needed to induce a producer to produce and sell each additional unit of a good is the marginal cost. This minimum amount is also called the ***reservation price***. The producer will produce and sell one more unit only if the market price is greater than or equal to this reservation price. For example, if the market price of a good is $20 while the reservation price is $10, then the producer surplus of that particular unit of good is $20 - $10 = $10. If he or she produces and sells more than one unit of the good, total producer surplus is the summation of the producer surplus he or she gains from each of the units sold at the market price.

Example 2 under the section Something to Think About on p.10 of Unit 1 can be a good illustration. If the columns of total cost and marginal cost describe your case, how many hours of tuition service you will provide per week and how much will be your producer surplus? The following table is copied from that example and includes three more columns to illustrate how the producer surplus (PS) can be derived and your rational decision.

| Hours per week | Marginal benefit ($) | Total cost ($) | Marginal cost ($) | Increase in PS ($) | Total PS ($) | Sell or not? |
|---|---|---|---|---|---|---|
| 0 | - | 0 | - | - | - | - |
| 1 | 200 | 30 | 30 | 170 | 170 | √ |
| 2 | 200 | 100 | 70 | 130 | 300 | √ |
| 3 | 200 | 220 | 120 | 80 | 380 | √ |
| 4 | 200 | 400 | 180 | 20 | 400 | √ |
| 5 | 200 | 650 | 250 | -50 | 350 | × |

If you understand the explanation of Example 1, you will figure out very easily this example is also a *discrete* case. The third last column in the table shows how much economic surplus you can gain from providing each additional hour of tuition service. The second last column shows the total producer surplus you can derive from providing a certain number of hours of such service. At the price of $200 per hour, you will only supply 4 hours of tuition service per week to maximize your producer surplus, which is equal to $400 in this case. Working longer hours will make you worse off, as shown by the decrease in producer surplus.

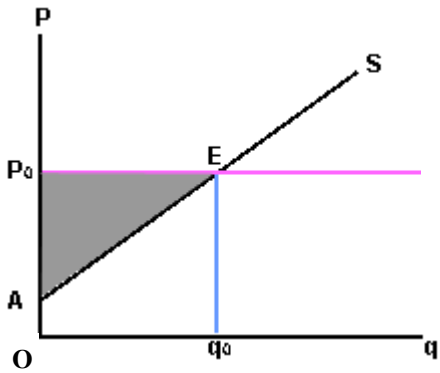



**Figure 2.5**
Producer Surplus

In the *continuous* case in which the quantity of a good is assumed to be perfectly divisible, a diagram such as Figure 2.5 will be useful to illustrate producer surplus. To understand the diagram, refer to Topic 1.3.4 that introduced two ways of interpreting the supply curve. The second interpretation is that the supply curve gives us the minimum price that a producer is willing to accept for a given quantity of output. Therefore, the area under the supply curve, given by the area **$OAEq_0$**, represents the minimum amount that makes the producer willing to produce **$q_0$** of the good. It also reflects the opportunity cost he has to incur for the production of that quantity. The actual amount that the producer receives is given by the area **$OP_0Eq_0$**. The difference between these two areas is the producer surplus, represented by the shaded area **$AP_0E$**. It is the net gain the producer can derive from the production and sale of the good.

If the supply curve in Figure 2.5 denotes the market supply, then the producer surplus gained by all the producers who sell the good at the market price $P_0$ can be illustrated in the same way as explained above.

# 2.3: Price Ceiling and Price Floor

In this section we study the two cases in which the market is prevented from reaching its equilibrium. This occurs when the government imposes either a price ceiling or price floor on the good.

## 2.3.1: Effect of Price Ceiling

Price ceiling is a regulation on the market price. It becomes illegal to buy and sell the good concerned above the regulated price. Rent ceiling is an example. A price ceiling is effective only if it is set below the equilibrium price.

**Something to Think About**

Regulated Rent of Public Housing Units in Hong Kong



Consider the rents of the public housing units managed by the Housing Authority in Hong Kong. Although the rental rate is fixed in both directions, it is in effect a price ceiling as it is lower than the market rental rate of comparative units in the private sector. This real life example is a good illustration of the effect of price ceiling. With reference to this example, can you explain why a price ceiling is effective only if it is set below the equilibrium price?

What will happen if an effective price ceiling is imposed on a good? It can be illustrated with Figure 2.6.

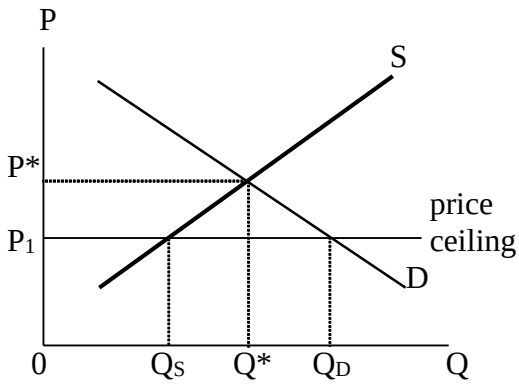



**Figure 2.6**
Effect of Price Ceiling

When a price ceiling prevents the market price from rising to its equilibrium level, there will be two phenomena in the market.

Persistent disequilibrium

> When a price ceiling $\mathbf{P_1}$ is set below the equilibrium **P***, there will be an **excess demand** represented by the difference between $\mathbf{Q_D}$ and $\mathbf{Q_S}$. As upward adjustment in price is not possible in this case, equilibrium cannot be restored. Disequilibrium will persist. The good in short supply will be allocated to the consumers by means of queuing, rationing, or first come first serve method. In the case of public housing units in Hong Kong, queuing and rationing according to certain criteria laid down by the Housing Authority are ways of allocating the units.

Winners and losers

> Why does the government impose a price ceiling that prevents the market from clearing? The apparent objective is to help one group of the economic agents who participate in the market transactions. Which group – buyers or sellers? Who will benefit from the effective price ceiling that is set below the

equilibrium price? Of course the buyers! But note that only those buyers who manage to buy the good or service at the price ceiling are winners. The sellers who are forced to sell at a lower price and those buyers who cannot buy that good are losers.

Black market

At the price ceiling $\mathbf{P_1}$, only $\mathbf{Q_S}$ will be offered for sale. Some of those buyers who are unable to buy the good at the regulated price may offer a higher price for the good and some producers may take the risk to sell to these consumers to gain a higher return. A black market develops. In San Francisco where there is a strict enforcement of rent ceiling, some landlords charge higher rent than the regulated level. But these black market traders will be punished once they are detected by the government.

## 2.3.2: Effect of Price Floor

Price floor is the opposite case of price ceiling. The imposition of a price floor makes it illegal to buy and sell the good concerned below the regulated price. A long-existing example in Hong Kong is the minimum wage applied to domestic helpers. (Do you know what is their current minimum wage per month?) A price floor is effective only if it is set above the equilibrium price.

The effect of an effective price floor can be illustrated with Figure 2.7.

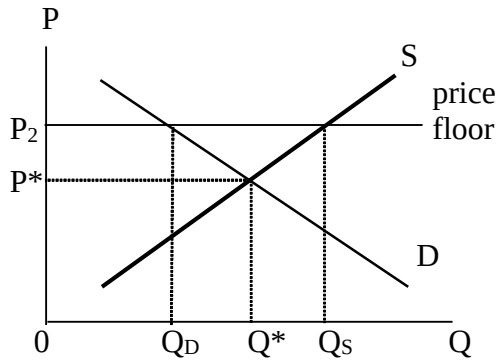



**Figure 2.7**
Effect of Price Floor

When a price floor prevents the market price from falling to its equilibrium level, there will be two phenomena in the market.

Persistent disequilibrium

When a price floor $\mathbf{P_2}$ is set above the equilibrium **P***, there will be an **excess supply** represented by the difference between $\mathbf{Q_S}$ and $\mathbf{Q_D}$. As downward adjustment in price is not possible in this case, equilibrium cannot be restored. Disequilibrium will persist. Stocks of unsold goods will be accumulated. In the case of a minimum wage, not all workers can find a job. There will be unemployment.

Winners and losers

Repeating the same questions as for price ceiling, we get the following answers. The apparent objective of imposing a price floor above the equilibrium price is to help the sellers of the good concerned. But note that only those sellers who manage to sell the good or service at the price floor are winners. The buyers who have to pay a higher price and those sellers who cannot sell at the price floor are losers.

Black market

At the price floor $\mathbf{P_2}$, consumers are only willing to buy $\mathbf{Q_D}$ of the good. Some of those sellers who are unable to sell their good at the regulated price may offer to reduce their price. A black market emerges. It is not uncommon for some of those domestic helpers who come to Hong Kong from India and Sri Lanka to accept a wage below the official minimum wage.

**Something to Think About**

Statutory Minimum Wage in Hong Kong



After years of heated debates in Hong Kong between unionists and employers and within the Legislative Council, the Statutory Minimum Wage (SMW) has finally come into force since 1 May 2011. The SMW rate was first set at $28 per hour, raised to $30 since 1 May 2013 and further to $32.5 with effect from 1 May 2015. The argument for enforcing minimum wage rates for local workers in Hong Kong is to protect their welfare by preventing employers from paying them very low wages. Do you think it is an effective way to help the unskilled workers?

The minimum monthly wage for domestic helpers has been enforced in Hong Kong for a long time. Is the objective of this policy to protect the welfare of the foreign helpers?

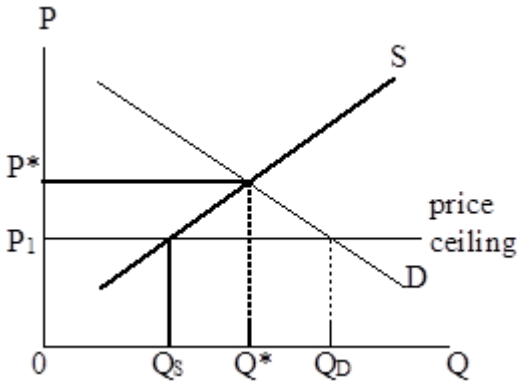
Now try the third self exercise.

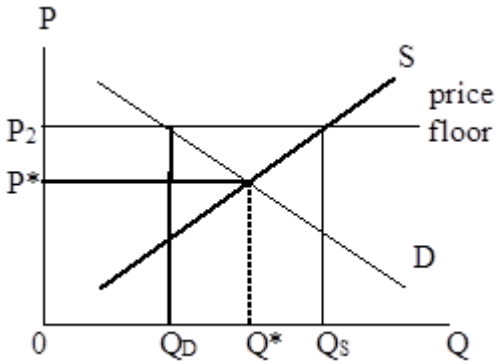
**Self Exercise 3**

Consumer and Producer Surplus under Price regulation



Show the areas that represent consumer surplus and producer surplus in each of the diagrams of effective price ceiling and effective price floor respectively. Can your answers show who are the winners and who are the losers in each case?




A response to this self exercise is on page 33.

# 2.4: Elasticity of Demand

In this topic we introduce the following elasticities of demand:

- price elasticity;
- income elasticity; and
- cross elasticity.

We will also analyse the implication of price elasticity of demand on the total revenue earned by the supplier of a good and the tax revenue collected by the government.

In Unit 1 and in Topic 2.1 of this unit we have studied the directional impacts of various determinants of demand and supply and how their changes will affect the market equilibrium. **In the rest of this unit we will focus only on demand**. In particular, we want to know if there will be a great or small change in the quantity demanded of a good following a change in its price, in consumers' income, or in the price of a related good.

# Responses to Self Exercises

This section contains the responses to the seven (7) self exercises in unit 2 – Market Equilibrium and Elasticity of Demand. Make sure you have attempted the exercises before you look at their responses.

**Response to Self Exercise 1**

Adjustment of Market Equilibrium

| Event | Price | | | Quantity | | |
|---|---|---|---|---|---|---|
| | **Increase** | **Decrease** | **Uncertain** | **Increase** | **Decrease** | **Uncertain** |
| 1. The increase in tariff (an import tax) rate on the imports of **wine**. | ✓ | | | | ✓ | |
| 2. The previously illegal sales of a **drug** are now legalised. | | | ✓ | ✓ | | |
| 3. Medical research has found that **kiwifruit** is a cure of cancer. | ✓ | | | ✓ | | |
| 4. How would a good harvest of potatoes affect the market of **French fries**? | | ✓ | | ✓ | | |
| 5. Over time the technology of computer production has improved substantially while the use of **computers** has become more common. | | | ✓ | ✓ | | |

Managed to tick all the right boxes? Well done!

**Event 1** leads to a decrease in supply (illustrated by a leftward or upward shift in the supply curve) as the imported wine will be sold at a higher price to cover the increased tariffs. It will raise the equilibrium price.

Both **event 2** and **event 5** are a little trickier than the others as they cause both demand and supply to shift to the right. In the case of the previously illegal drug, once its sales are legalised, there will not be any penalties (fines or imprisonment) and hence the costs will be lower and the supply will increase. The demand will increase for the same reason. When both demand and supply increase, the quantity transacted will definitely increase. But the change in equilibrium price is uncertain based on theoretical analysis. It depends on which of the two curves shifts more. Therefore you are correct to select "Uncertain" as the answer for equilibrium price.

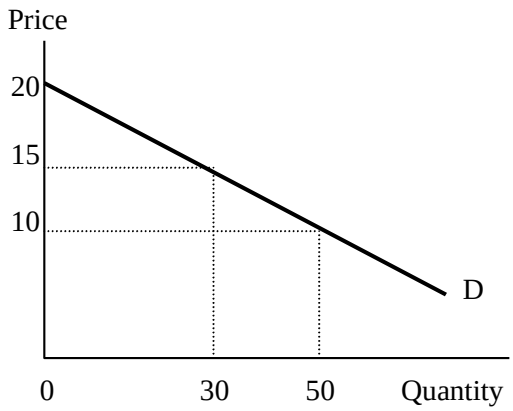
**Event 3** will raise the demand for kiwifruit leading to an increase in both equilibrium price and quantity.

**Event 4** will increase the supply of french fries because a good harvest of potatoes will lower the price of this crucial raw material and reduce the cost of production. Given an unchanged demand for french fries, the equilibrium price will drop and equilibrium quantity will rise.

## Response to Self Exercise 2

### Consumer Surplus

I hope you can manage the calculations required. The answers are given below.




1. When the price equals 15, what is the maximum amount that David is willing to pay for the 30 units of X?

   **525**

   The maximum amount that David would be willing to pay for the 30 units of X is given by the area under the demand curve up to the quantity of 30 units. The area of that trapezoid is equal to $(15 + 20) \times 30 \div 2$.

2. What is David's consumer surplus when the price equals 15?

**75**

By definition, David's consumer surplus is the excess of the amount that he is willing to pay over the actual amount he pays for the good. When the price is 15, he pays 450 for the 30 units of X. Therefore, his consumer surplus is 525 – 450 = 75. Another easier way is simply to work out the area of the triangle above the price of 15.

3. What is David's consumer surplus when the price is reduced to 10?

**250**

The consumer surplus in this case is equal to the area of the triangle above the price of 10. That is, (20–10) ×50÷ 2 = 250.

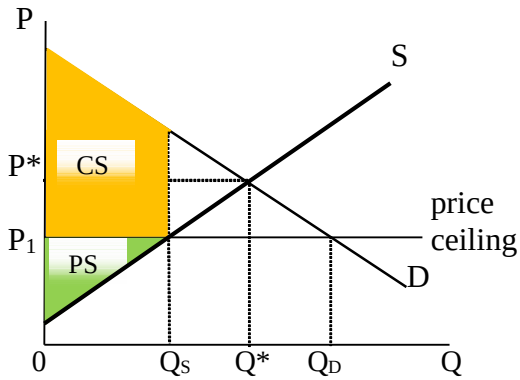
4. What can you conclude from the above analysis?

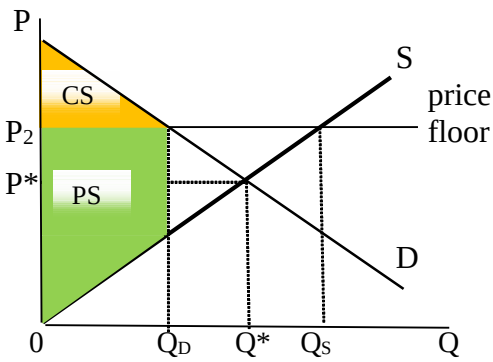
The lower the price of a good, the **larger** the consumer surplus.

**Response to Self Exercise 3**

Price Ceiling and Price Floor

Here is the answer.




In the two diagrams, the orange areas depict consumer surplus (CS) and the green areas depict the producer surplus (PS).

Excellent if you have managed to correctly identify the areas of consumer surplus and producer surplus!

It is very obvious that CS increases and PS decreases after the imposition of an effective price ceiling. Therefore, consumers as a group are the winner while producers are losers.

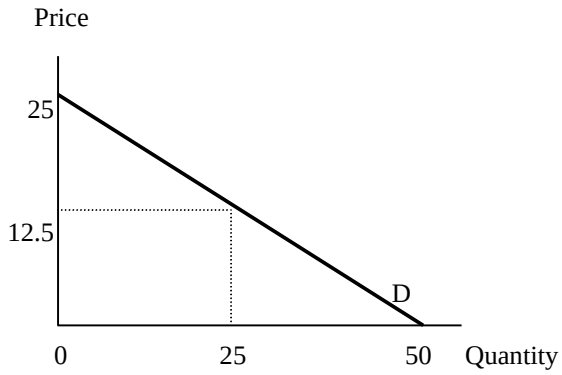
In the case of an effective price floor, the areas indicate that producers as a group are winner while consumers are losers.

Later on in Topic 4 you will learn the concept of deadweight loss, which is the reduction in social surplus relative to its maximum possible value. You will realize then that both price ceiling and price floor create deadweight loss.

**Response to Self Exercise 4**

Price Elasticity and Total Revenue

The answers are given with reference to the diagram below.




1. When the price is 20, is the demand elastic or inelastic?
   **elastic** - as 20 is higher than the price of 12.5 at the mid-point of the demand curve, demand is elastic.

2. When the price is 10, is the demand elastic or inelastic?
   **inelastic** - at the price of 10, demand is at the lower half of the demand curve and therefore is inelastic.

3. At what quantity is the total revenue of the firm maximised?
   **25** - total revenue is maximised at the point of unit elasticity. That is, the mid-point of the demand curve.

4. What is the value of the maximum total revenue?
   **312.5** - this is given by price × quantity. That is, 12.5 × 25.

5. What is the value of the price elasticity of demand when total revenue is maximised?
   **1** - as explained in the notes.